PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Artha AI

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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CONTENTS

S.No.	Title	Page No.
	Acknowledgement	4
	List of Figures and Tables	5
	Abstract	6
1	Introduction	7
2	Methodology	8
3	Tools and Technologies	9
4	Implementation	10
5	Major Findings/Outcomes/Output/Results	11
6	Conclusion and Future Scope	11
	References	12
	Appendices	13

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Artha AI"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gifty Gupta**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Date: ……..

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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LIST OF FIGURES

S. No.	Title	Page No.
A.1	Landing Page	13
A.2	Dashboard	14
A.3	Expense Management	14
A.4	Goal Planning	15
A.5	Recommendations Module	16

ABSTRACT

Within the contemporary fiscal environment, practitioners and laypersons alike encounter significant hurdles regarding the systematic oversight of personal earnings, expenditures, and enduring monetary objectives. Current digital offerings frequently prioritize disparate functions, thereby neglecting to offer a holistic framework for comprehensive financial governance

.

The **Artha AI** initiative represents an intelligent advisory architecture engineered to consolidate budget monitoring, strategic planning, and automated fiscal evaluation. By leveraging sophisticated rule-based methodologies, the application scrutinizes transactional patterns to produce substantive, data-driven guidance for the user.

This infrastructure utilizes a robust client-server model—incorporating React, Node.js, and MongoDB—to facilitate modular scalability and operational performance. Ultimately, the project endeavors to enhance economic literacy, empowering individuals to navigate complex financial choices through rigorous computational analysis and tailored strategic insights.

1. Introduction

1.1. Background

Financial management is a critical aspect of everyday life, especially for students and young professionals. However, many individuals lack access to tools that provide structured financial guidance.Existing financial applications :

- Focus only on expense tracking or investment monitoring
- Lack integration of financial planning features
- Do not provide intelligent recommendations

This creates a gap in effective financial decision-making.

1.2. Problem Statement

Users face difficulties in :

- Managing expenses efficiently
- Planning financial goals
- Understanding savings behavior
- Navigating complex financial systems

There is a need for a unified system that integrates all financial aspects.

1.3. Objective

The objective of ArthaAI is to:

- Track income and expenses
- Analyze financial behavior
- Provide goal-based planning
- Generate intelligent financial recommendations

1.4. Scope of the Project

The system focuses on:

- Personal finance management
- Expense tracking
- Goal planning
- Financial recommendation generation

It is designed primarily for Indian users, considering financial behavior and requirements.

2. Methodology

2.1. Development Approach

The project follows a modular and layered approach :

- Requirement Analysis
- System Design
- Backend Development
- Frontend Development
- Integration
- Testing

2.2. System Design

The system is divided into layers :

- Client Layer
- API Layer
- Business Logic Layer
- Recommendation Engine
- Database Layer

2.3. Workflow

- User inputs financial data
- Data is processed by backend logic
- Recommendation engine analyzes patterns
- Results are displayed on dashboard

3. Tools and Technologies

3.1. Frontend

- React.js
- Tailwind CSS
- Recharts
- Framer Motion

3.2. Backend

- Node.js
- Express.js

3.3. Database

- MongoDB (NoSQL)

3.4. Authentication

- JSON Web Tokens (JWT)

3.5. Development Tools

- VS Code
- Postman
- Git & GitHub

4. Implementation

4.1. Frontend Implementation

- Built using React with component-based architecture
- Reusable components (Button, Card, Input)
- Dashboard with charts and analytics

4.2. Backend Implementation

- RESTful APIs using Express
- Controllers for handling business logic
- Middleware for authentication

4.3. Database Implementation

Collections :

- Users
- Expenses
- Goals

Flexible document-based schema.

4.4. Recommendation Engine

1. Rule-based system

2. Evaluates :

    - Savings ratio
    - Spending patterns
    - Risk profile

4.5. UI/UX Design

1. Clean dashboard layout

2. Light theme interface

3. Responsive design

5. Major Findings / Results

- Users gain better understanding of their spending patterns
- System helps identify overspending
- Goal tracking improves financial discipline
- Recommendation engine provides actionable insights
- Observations :

  - Modular architecture improves scalability
  - MongoDB provides flexibility in handling financial data
  - Real-time analytics enhances user experience

6. Conclusion and Future Scope

6.1. Conclusion

ArthaAI successfully demonstrates a complete personal finance management system that integrates multiple functionalities into a single platform.

The project highlights :

- Full-stack development skills
- Scalable system design
- Practical application of financial logic

6.2. Future Scope

- Tax optimization system
- AI/ML-based predictions
- Investment recommendation module
- Mobile application
- Notification system

REFERENCES

1. MongoDB Documentation
   (https://www.mongodb.com/docs/)

2. React Documentation
   (https://react.dev/learn)

3. Node.js Documentation
   (https://nodejs.org/docs/latest/api/)

4. Express.js Documentation
   (https://expressjs.com/en/starter/installing.html)

APPENDICES

1. Appendix A : User Interface Screenshots

This section contains additional screenshots of the ArthaAI application to provide a visual understanding of the system.

A.1 Landing Page

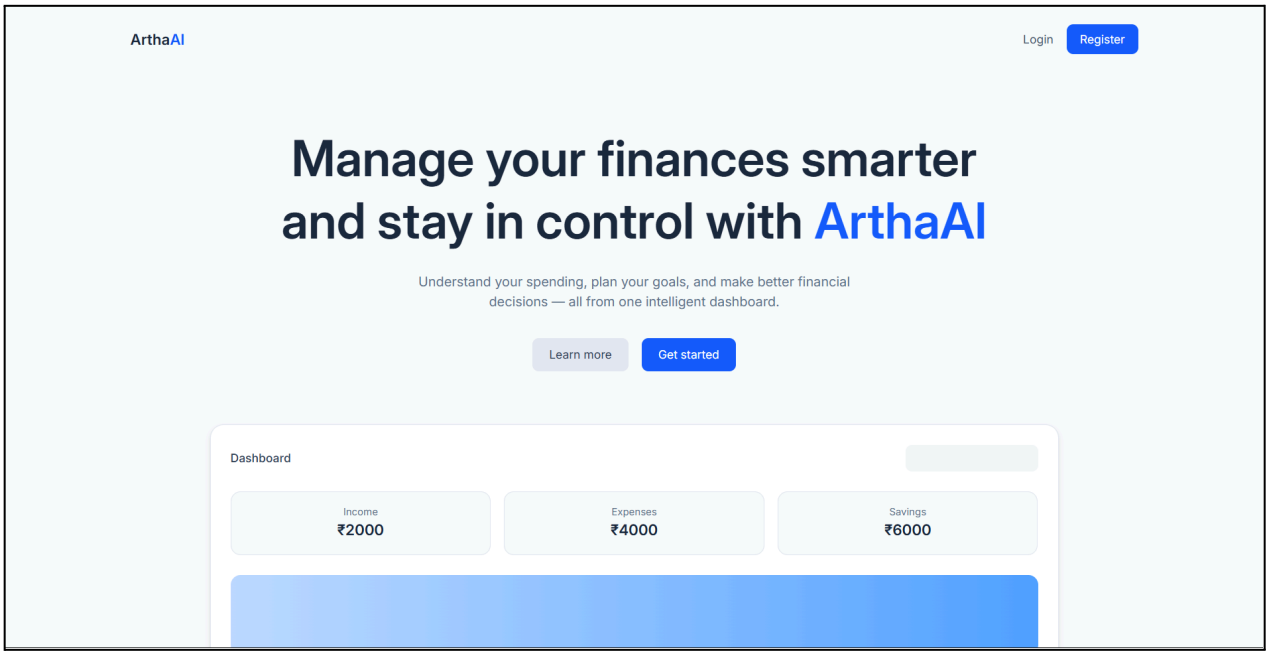

Figure A1: Landing page of ArthaAI

The landing page introduces the platform and highlights its key features. It serves as the entry point for users to register or log in.

A.2 Dashboard

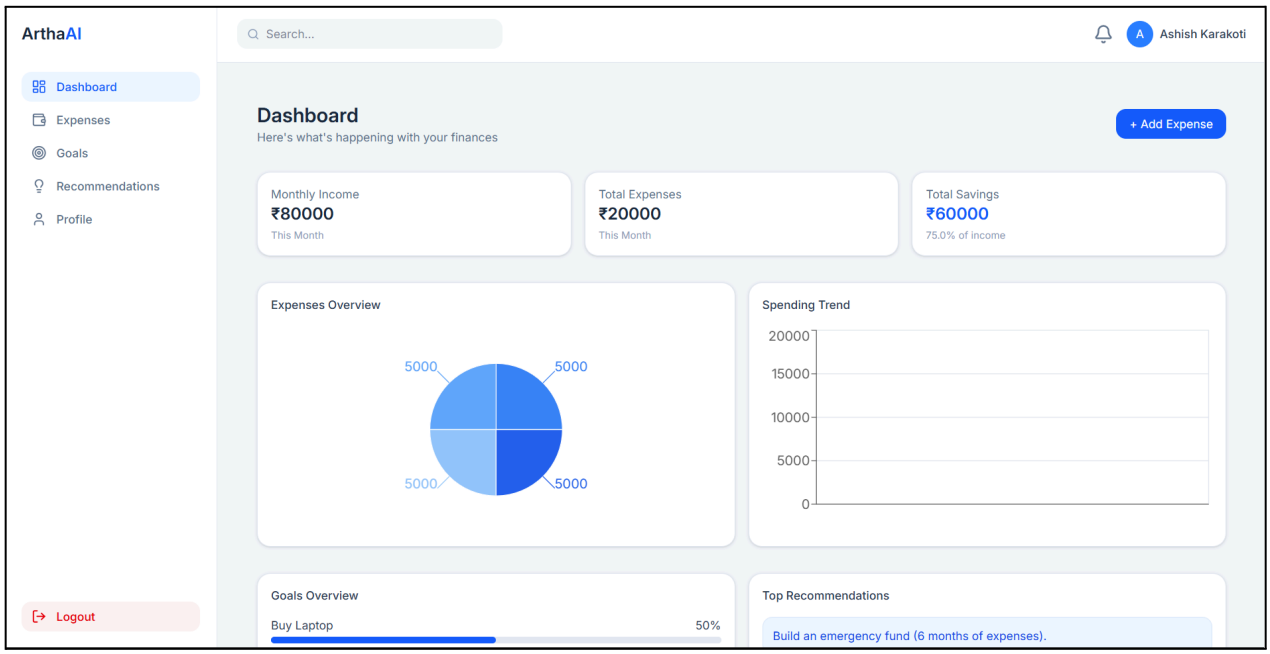

Figure A2: Dashboard interface

Displays financial summary including income, expenses, savings, and graphical insights for better analysis.

A.3 Expense Management

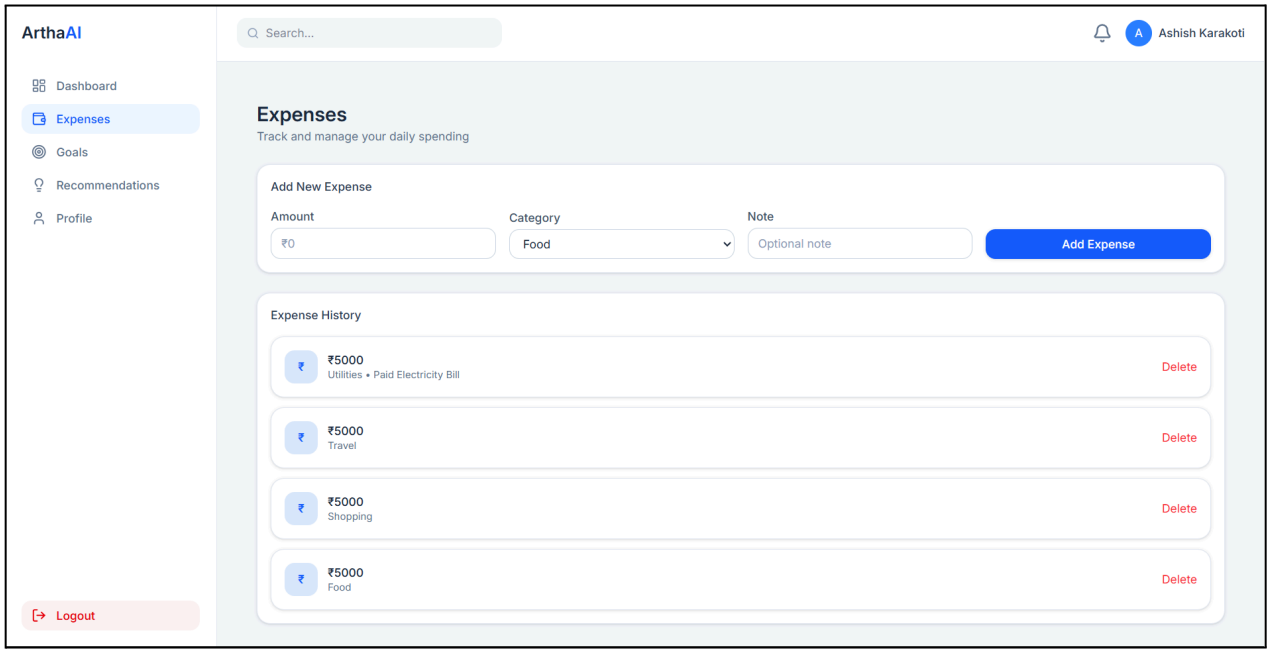

Figure A3: Expense tracking module

Allows users to add, view, and manage categorized expenses efficiently.

A.4 Goal Planning

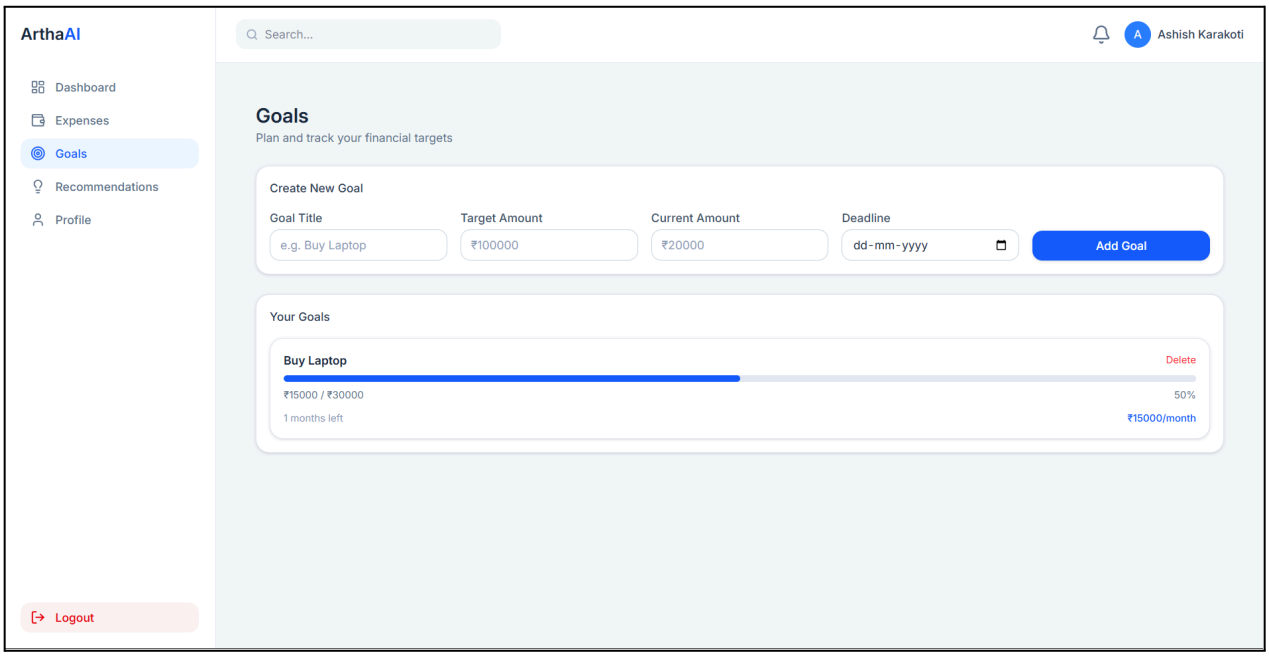

Figure A4: Goal planning module

Shows financial goals with progress tracking and required monthly contribution.

A.5 Recommendations Module

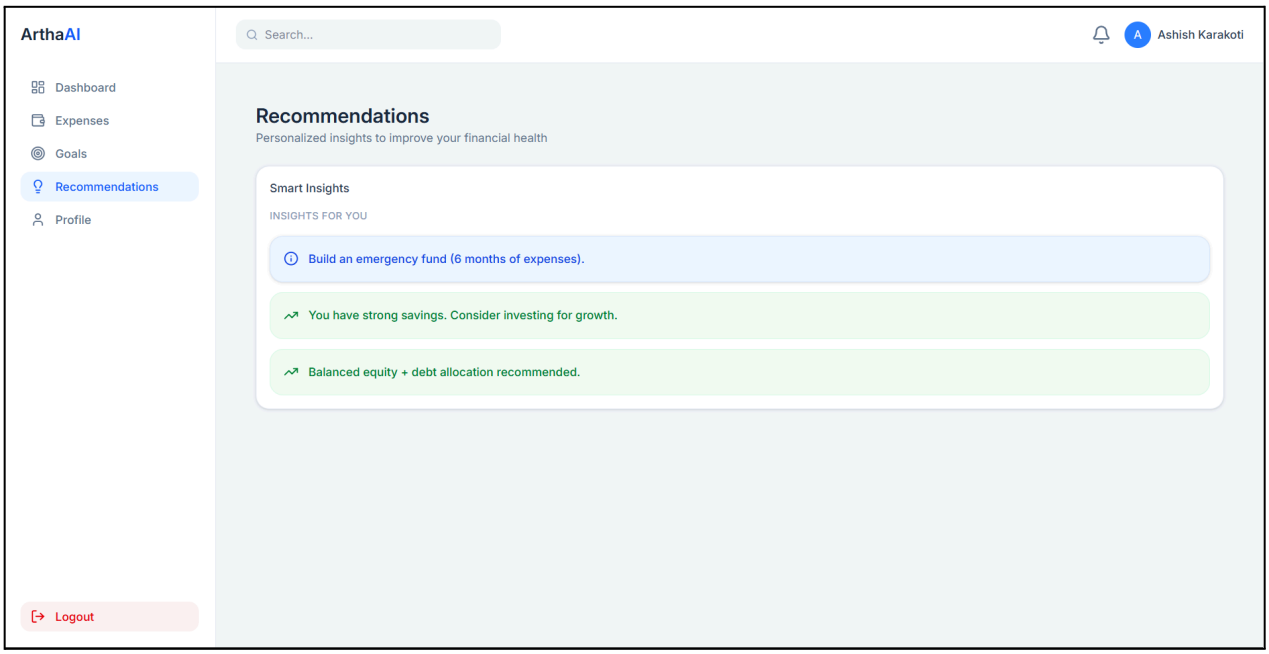

Figure A5: Recommendation system interface

Displays personalized financial insights based on user behavior.

2. Appendix B : API Endpoints

The backend provides RESTful APIs for handling application functionality.

Method	Endpoint	Description
POST	/auth/register	Register new user
POST	/auth/login	User login
GET	/user/profile	Get user profile
PUT	/user/profile	Update profile
GET	/expenses	Get all expenses
POST	/expenses	Add expense
DELETE	/expenses/:id	Delete expense
GET	/goals	Get goals
POST	/goals	Add goal
DELETE	/goals/:id	Delete goal
GET	/recommendations	Get recommendations

---

3. Appendix C : Database Schema

The system uses MongoDB with the following collections :

a. Users Collection

- Name
- Email
- Password
- Monthly Income
- Risk Profile

---

b. Expenses Collection

- User ID
- Amount

- Category
- Note
- Date

---

c. **Goals Collection**
    - User ID
    - Title
    - Target Amount
    - Current Amount
    - Deadline

---

4. Appendix D : Sample Data

Example expense record :

```
{
  "amount": 500,
  "category": "food",
  "note": "Lunch",
  "date": "2026-04-01"
}
```

---

5. Appendix E : Additional Notes

- The system is designed using a modular architecture
- Backend logic is rule-based for transparency
- The application can be extended with AI/ML models in future